

BIOLOGICAL DATABASES (BioDB)

Comprehensive Lab Internal & Quiz Master Preparation Handbook
Covers Modules 1, 3, 4, 5, 6 | In-depth Theory + Step-by-Step Practical Protocols

Module	Focus Areas & Topics Covered	Practical / Lab Skills
Mod 1: Sequence Submission	Central dogma, RDBMS, BankIt, Webin, SAKURA, FASTA, GenBank, EMBL, ReadSeq.	File format syntax, header parsing, sequence format conversion via ReadSeq/seqret.
Mod 3: Seq Databases	INSDC (GenBank, EMBL-EBI, DDBJ), UniProt (Swiss-Prot vs TrEMBL), PIR, COSMIC, ClinVar, dbSNP.	RefSeq vs GenBank IDs, rsID querying, filtering clinically significant variants.
Mod 4: Structure DBs	PDB, mmCIF, coordinate records, SCOP (C-F-S-F), CATH (C-A-T-H).	PDB header parsing, ATOM vs HETATM extraction, calculating resolution, domain classification.
Mod 5: Function & Pathways	Pfam (HMMs), PROSITE patterns/profiles, Gene Ontology (BP, MF, CC), EC classes, KEGG, BioGRID, STRING, DIP.	Writing PROSITE regex, interpreting KEGG wiring diagrams, constructing PPI networks & confidence scores.
Mod 6: Genome & Microarray	Ensembl, UCSC Genome Browser (Tracks, BLAT), DNA Microarray (Cy3/Cy5), GEO (GSM, GSE, GPL, GDS), SAGE.	UCSC BLAT genomic alignment, track visual navigation, GEO accession retrieval & probe mapping.

PART 1: COMPREHENSIVE THEORY & QUIZ PREPARATION

1. Module 1: Sequence Submission Tools & Molecular Data Formats

A. Motivation of Biological Databases & RDBMS: Biological data generated by next-generation sequencing (NGS), mass spectrometry, and X-ray crystallography is massive, heterogeneous, and exponentially growing. Biological databases store, organize, index, and retrieve biological data systematically. A **Relational Database Management System (RDBMS)** organizes data into tables (relations) with predefined rows (tuples/records) and columns (attributes). Tables are interconnected through *Primary Keys* (unique identifiers within a table) and *Foreign Keys* (references to primary keys in another table). Relational algebra allows ACID compliance (Atomicity, Consistency, Isolation, Durability) and powerful SQL queries.

B. Central Dogma of Molecular Biology: Proposed by Francis Crick (1958). DNA undergoes semi-conservative *Replication*, is transcribed into mRNA by RNA Polymerase (*Transcription*), and mRNA codons (triplets) are translated into polypeptide chains on ribosomes by tRNA (*Translation*). Reverse transcription (RNA $\rightarrow$ DNA) occurs in retroviruses via Reverse Transcriptase.

C. International Nucleotide Sequence Collaboration (INSDC): Composed of GenBank (NCBI, USA), ENA/EMBL-Bank (EMBL-EBI, Europe), and DDBJ (NIG, Japan). They exchange newly submitted nucleotide data on a daily synchronized basis. Once an accession number is granted by any of the three, it is globally recognized and universal.

D. Sequence Submission Portals:

- **BankIt (NCBI):** Web-based questionnaire wizard. Designed for simple submissions: single/few mRNA, genomic DNA, organellar DNA, ribosomal RNA (16S/18S/ITS), or viral sequences. No local software installation needed.
- **Sequin (NCBI):** Standalone graphical program developed by NCBI for complex, long, or multi-segmented genomes. (Note: Officially retired by NCBI in favor of the web Submission Portal, but frequently tested in exams).
- **Webin (EMBL-EBI):** Standard submission system for nucleotide sequences, raw reads (SRA), assembled genomes, and transcriptomes to ENA.
- **SAKURA / D-way (DDBJ):** Web-based data submission system operated by the National Institute of Genetics (NIG) in Japan.

E. Molecular Sequence Formats:

- **FASTA (.fasta / .fa):** Starts with '>' line followed by sequence ID and optional description. Subsequent lines contain pure 1-letter codes. Line breaks typically occur at 60 or 80 characters.
- **GenBank Flat File (.gb / .gbk):** Rich text record with labeled tokens: LOCUS (length, molecule, division, date), DEFINITION, ACCESSION, VERSION (includes GI number historically), DBASE, SOURCE, REFERENCE, FEATURES (CDS, gene, exon coordinates, /translation), ORIGIN (numbered base sequence ending in '//').

- **EMBL Format (.dat):** Uses two-character line codes: ID (identification), AC (accession), SV (sequence version), DE (description), KW (keywords), OS (organism species), OC (organism classification), FH/FT (feature header/table), SQ (sequence header), and terminates with '//'.
- **PIR / NBRF:** Starts with '>P1;' or '>F1;' followed by identifier, second line is description, sequence ends with an asterisk '*'.
- **Format Interconversion:** Carried out by web tools like **ReadSeq** (Don Gilbert) or command-line EMBOSS utility **seqret**.

2. Module 3: Nucleotide and Protein Sequence Databases

A. Primary vs. Secondary Databases: Primary databases contain raw archival experimental submissions without manual curation (GenBank, ENA, DDBJ, PDB). Secondary databases are derived, curated, non-redundant, and computationally or expert-annotated (RefSeq, Swiss-Prot, Pfam, PROSITE).

B. UniProt Knowledgebase (UniProtKB): The central hub for functional protein information. Formed by the UniProt Consortium (EMBL-EBI, SIB, and PIR). Consists of two sections:

- **UniProtKB/Swiss-Prot:** Manually annotated and reviewed by expert biocurators. Features high-quality functional descriptions, domain boundaries, post-translational modifications (PTMs), variants, and literature citations. Extremely low redundancy.
- **UniProtKB/TrEMBL:** (Translated EMBL). High-throughput, automatically translated and computationally annotated from open reading frames (CDS) of ENA/GenBank/DDBJ. Not yet reviewed by human curators.
- **PIR (Protein Information Resource):** Founded by Margaret Dayhoff (pioneer of bioinformatics who created the PAM matrix). Developed the Protein Sequence Database (PIR-PSD), now integrated into UniProt.

C. Genetic Disorders & Mutation Databases:

- **COSMIC (Catalogue Of Somatic Mutations In Cancer):** Curated by the Wellcome Sanger Institute. World's most comprehensive resource for somatic mutation data in human cancers (point mutations, fusions, copy-number variants, drug resistance).
- **ClinVar (NCBI):** Public archive of relationships between human genomic variations and observed clinical phenotypes, with supporting evidence. Variants are categorized as: *Pathogenic*, *Likely Pathogenic*, *Uncertain Significance (VUS)*, *Likely Benign*, or *Benign*.
- **dbSNP (NCBI):** Repository for short genetic variations (single nucleotide variations, short deletions/insertions, microsatellites). Assigns a permanent reference SNP cluster ID: **rsID** (e.g., 'rs1801133'). Submitter-level records have 'ssIDs' before being clustered into an 'rsID'.

3. Module 4: Protein Structure Databases & Classification

A. Protein Data Bank (PDB): Worldwide repository for 3D macromolecular structures of proteins and nucleic acids, managed by the **wwPDB** consortium (RCSB PDB, PDBe, PDBj, BMRB).

- **Experimental Methods:** X-ray Crystallography (~85% of PDB; measures electron density, reports atomic resolution in Ångströms; $\leq 2.0 \text{ Å}$ is considered high resolution); Nuclear Magnetic Resonance (NMR; captures ensemble of conformations in solution, reports distance constraints); Cryo-Electron Microscopy (Cryo-EM; high-resolution for large complexes without crystallization).
- **PDB Format Structure:** 80-column fixed-width text records. Key record types: 'HEADER', 'COMPND', 'REMARK 2' (Resolution), 'SEQRES' (Primary sequence), 'ATOM' (Standard coordinates of protein/nucleic acid atoms), 'HETATM' (Hetero-atoms: water, ions, ligands, inhibitors), 'CONECT' (Bond connections), 'TER' (End of chain).
- **mmCIF (macromolecular Crystallographic Information File):** Modern replacement for PDB flat files that eliminates limitations on chain count (>62 chains) and atom count (>99,999 atoms).

B. Hierarchical Structural Classification Systems:

Proteins share structural folds even when sequence similarity is undetectable (<20% identity). Two premier classification databases:

Feature	SCOP (Structural Classification of Proteins)	CATH (Class, Architecture, Topology, Homology)
Founders & Origin	Alexey Murzin et al. (MRC LMB, Cambridge, 1995).	Christine Orengo et al. (UCL, London, 1997).
Methodology	Largely manual / visual inspection by experts.	Semi-automated: algorithms (SSAP) + expert curation.
Hierarchy Levels	Class: Overall secondary structure composition. Fold: Common structural core arrangements. Superfamily: Probable common evolutionary origin. Family: Clear evolutionary relationship (>30% identity).	C (Class): Secondary structure composition. A (Architecture): Spatial orientation, no connectivity. T (Topology): Fold & connectivity of secondary elements. H (Homologous Superfamily): Common ancestry.
Unique Level	Fold: Groups proteins with similar secondary structure topologies regardless of evolutionary relationship.	Architecture: Describes overall shape (e.g., α - β barrel, sandwich, roll) independent of loop connectivity.

4. Module 5: Protein Function, Pathway and Interaction Databases

A. Pfam (Protein Families Database): Comprehensive collection of protein domains and families represented by **Hidden Markov Models (Profile HMMs)** generated using the 'HMMER' software package. Pfam consists of two tiers: **Pfam-A** (high-quality, manually curated seed alignments and curated thresholds) and **Pfam-B** (automated, uncurated clusters from ADDA).

B. PROSITE (Signatures and Profiles): Identifies functional sites, motifs, and domains using two methods:

- **Patterns:** Strict regular expressions representing conserved active sites. Rules: '[AC]' means Ala or Cys; '{ED}' means neither Glu nor Asp; 'x' means any amino acid; 'x(4)' means four arbitrary residues; 'x(2,4)' means 2 to 4 residues; '<' means N-terminal; '>' means C-terminal. Example: '[LIVM]-x(2)-[LIVM]-x(2)-[LIVM]'.
- **Profiles:** Position-Specific Scoring Matrices (PSSMs) incorporating insertion and deletion penalties to model divergent domain families.

C. Gene Ontology (GO): Provides a standardized, controlled, species-independent vocabulary for gene and protein attributes. GO is structured as a **Directed Acyclic Graph (DAG)** where terms can have multiple parents. Three strictly defined organizing domains:

- **Molecular Function (MF):** Elemental biochemical activities (e.g., 'catalytic activity', 'kinase activity', 'ATP binding').
- **Biological Process (BP):** Broad biological objectives executed by multiple molecular steps (e.g., 'signal transduction', 'cell division', 'glycolysis').
- **Cellular Component (CC):** Subcellular anatomy or macromolecular complexes where the gene acts (e.g., 'mitochondrial inner membrane', 'ribosome', 'nucleus').

D. ENZYME & Enzyme Commission (EC Numbers): Classification scheme based on reaction type, consisting of 4 numbers: 'EC a.b.c.d'.

- **EC 1: Oxidoreductases** (Oxidation-reduction reactions; dehydrogenases, oxidases).
- **EC 2: Transferases** (Transfer of chemical groups: methyl, phosphate; kinases, methyltransferases).
- **EC 3: Hydrolases** (Hydrolytic cleavage of bonds: esterases, nucleases, proteases).
- **EC 4: Lyases** (Non-hydrolytic/non-oxidative cleavage forming double bonds; decarboxylases).
- **EC 5: Isomerases** (Geometric or structural changes within a molecule; epimerases, mutases).
- **EC 6: Ligases** (Joining two molecules coupled with ATP hydrolysis; synthetases, DNA ligase).
- **EC 7: Translocases** (Movement of ions or molecules across membranes).

E. KEGG (Kyoto Encyclopedia of Genes and Genomes): Integrates genomic, chemical, and systemic functional information. Features **KEGG PATHWAY** (manually drawn wiring diagrams of metabolic and signaling networks). Each node represents a **KEGG Orthology (KO)** identifier (e.g., 'K00844' for Hexokinase).

F. Protein-Protein Interaction (PPI) Databases:

- **BioGRID:** Biomedical interaction repository curating physical interactions (Co-IP, Affinity Capture) and genetic interactions (synthetic lethality) directly from primary literature.
- **STRING (Search Tool for the Retrieval of Interacting Genes/Proteins):** Predicts physical and functional interactions. Integrates multi-evidence channels: *Gene Neighborhood*, *Gene Fusion*, *Co-occurrence*, *Co-expression*, *Experimental Assays*, *Curated Databases*, and *Automated Text Mining*. Assigns a combined confidence score (\$0.0 - 1.0\$; \$≥ 0.7\$ is high confidence).
- **DIP (Database of Interacting Proteins):** Curates experimentally determined physical protein interactions, subjected to automated and manual quality checks (DIP Core set).

5. Module 6: Genome and Microarray Databases

A. Genome Browsers: Interactive graphical representations of annotated reference genomes.

- **Ensembl (EMBL-EBI / Sanger):** Automated gene annotation pipeline for chordate/vertebrate genomes. Integrates comparative genomics, gene trees, regulatory features, and **BioMart** for batch filtering and export.
- **UCSC Genome Browser:** Highly responsive genome visualizer developed at UC Santa Cruz. Displays annotations as horizontal 'Tracks' (Genes, ESTs, SNPs, Repeats, Conservation, ENCODE regulation). Features **BLAT** (Blast-Like Alignment Tool; ultra-fast mRNA/DNA mapping) and **Table Browser** for querying tabular genomic coordinates.

B. DNA Microarray Principles & Technology:

Based on high-throughput microscopic nucleic acid hybridization between labeled target cDNA and known immobilized probe oligonucleotides on a solid substrate (glass chip).

- **Two-Color Microarray (cDNA):** Experimental sample labeled with Cy5 (Red, 650 nm) and Control sample labeled with Cy3 (Green, 550 nm). Competitive hybridization to the same array. Yellow = equal expression (\$Cy5/Cy3 = 1\$); Red = upregulated in target; Green = downregulated in target.

- **Single-Channel Microarray (Affymetrix GeneChip):** In situ synthesized oligonucleotide 25-mers. Each sample is hybridized to a separate chip and detected using biotinylated cRNA and streptavidin-phycoerythrin.

C. NCBI GEO (Gene Expression Omnibus): The premier international public repository for high-throughput functional genomics data (Microarray, RNA-seq, ChIP-seq).

• **GEO Accession Hierarchy (Extremely Important!):**

1. **GSM (GEO Sample):** Describes conditions, extraction, and quantitative expression values of an individual biological sample.
2. **GSE (GEO Series):** Group of related samples defining a complete study/experiment.
3. **GPL (GEO Platform):** Technical description of the array/sequencer (probe sequences, reporter IDs, manufacturer).
4. **GDS (GEO Dataset):** Curated, normalized, biologically comparable collections of GSMs.

D. SAGE (Serial Analysis of Gene Expression): Developed by Victor Velculescu and Bert Vogelstein (1995). Tag-based quantitative transcriptomics technique. Principles:

- mRNA is captured using oligo(dT) beads $\rightarrow$ synthesized to cDNA $\rightarrow$ cleaved with an anchoring restriction enzyme (e.g., *NlaIII*) $\rightarrow$ ligated to linkers containing a recognition site for a tagging enzyme (e.g., *BsmFI* or *MmeI*) $\rightarrow$ cleaves short 10–14 bp (or 21 bp in LongSAGE) sequence tags from a fixed position $\rightarrow$ tags are ligated into ditags $\rightarrow$ concatenated into long molecules $\rightarrow$ sequenced $\rightarrow$ tags are matched to known mRNA sequences.
- **Advantage over Microarray:** SAGE is open-ended (does not require prior knowledge of transcripts/probes; can discover novel genes) and provides digital absolute abundance counts. **Disadvantage:** Labor-intensive, costly, short tags can map ambiguously.

HIGH-YIELD QUIZ & MCQ PRACTICE QUESTIONS (WITH ANSWERS)

Q1: Which consortium synchronizes data daily between GenBank, ENA, and DDBJ?

Answer: INSDC (International Nucleotide Sequence Database Collaboration).

Q2: In UniProtKB, what is the primary difference between Swiss-Prot and TrEMBL?

Answer: Swiss-Prot is manually annotated and reviewed by biocurators; TrEMBL is computationally generated and unreviewed.

Q3: What does an 'rs' prefix indicate in human genetic variation?

Answer: Reference SNP cluster identifier in NCBI dbSNP (e.g., rs6025 for Factor V Leiden).

Q4: Name the 4 hierarchical classification levels of CATH in exact order.

Answer: Class (C), Architecture (A), Topology (T), Homologous superfamily (H).

Q5: In a PDB coordinate file, which record type holds solvent (water) and inhibitor molecules?

Answer: HETATM (Hetero atom). Standard amino acid/nucleic acid residues use ATOM.

Q6: What mathematical model forms the foundation of Pfam domain profiles?

Answer: Hidden Markov Models (Profile HMMs), implemented via HMMER.

Q7: In PROSITE syntax, what does the pattern '[LIVM]-x(2)-[DE]' mean?

Answer: A hydrophobic residue (Leu, Ile, Val, or Met), followed by any 2 amino acids, followed by an acidic residue (Asp or Glu).

Q8: What are the three non-overlapping organizing principles of Gene Ontology (GO)?

Answer: Molecular Function (MF), Biological Process (BP), and Cellular Component (CC).

Q9: What is the primary enzyme class for DNA Ligase in the EC system?

Answer: EC 6 (Ligases - enzymes that join two molecules coupled with hydrolysis of a pyrophosphate bond).

Q10: In STRING network graphs, what does a light-blue line connecting two nodes signify?

Answer: Curated database evidence (interaction verified from pathways or interaction databases).

Q11: What are the four main accession prefixes in NCBI GEO and what does each represent?

Answer: GSM = Sample, GSE = Series (Study), GPL = Platform (Array design), GDS = Curated Dataset.

Q12: In a two-color cDNA microarray, if a spot fluoresces bright green (Cy3), what does it mean?

Answer: The gene is downregulated in the experimental/target sample (higher expression in control labeled with Cy3).

Q13: Which database specifically catalogs somatic mutations found exclusively in human cancer tissues?

Answer: COSMIC (Catalogue Of Somatic Mutations In Cancer).

Q14: What is the specialized fast alignment tool integrated into the UCSC Genome Browser?

Answer: BLAT (Blast-Like Alignment Tool).

Q15: Why is SAGE considered a 'digital' gene expression method compared to microarrays?

Answer: SAGE counts absolute discrete frequency of sequenced tag occurrences, whereas microarray measures analog fluorescence intensity.

PART 2: LAB PRACTICAL PROTOCOLS, WORKFLOWS & VIVA TIPS

1. Practical Protocol: Molecular Sequence Formats & Conversions

Lab Task 1: Converting Sequence Formats using ReadSeq or EMBOS seqret:

In the lab exam, you will frequently be given a GenBank flat file and asked to convert it to FASTA, or vice versa, and isolate specific CDS regions.

Exact Command-Line Syntax (EMBOSS seqret):

```
# Convert GenBank flatfile to FASTA format:
seqret -sequence input.gbk -outseq output.fasta -osformat fasta

# Convert FASTA to EMBL format:
seqret -sequence input.fasta -outseq output.embl -osformat embl

# Extract only the coding sequence (CDS) translation:
# Parse the /translation="..." field directly from GenBank FEATURES table
```

Web-Based ReadSeq Execution Steps:

1. Open ReadSeq (e.g., EBI ReadSeq or Indiana Univ server).
2. Paste sequence into input text box or upload `.gbk` file.
3. Under 'Output Format', select 'FASTA / Pearson' or 'EMBL'.
4. Ensure 'Case' is preserved (Upper) and click 'Format' / 'Submit'.
5. Save the generated file with extension `.fasta`.

2. Practical Protocol: NCBI GenBank Submission via BankIt

Step-by-Step Practical Submission Workflow:

- **Step 1: Contact Information:** Enter submitter name, department, institution, country, and email.
- **Step 2: Reference Information:** Specify unpublished manuscript status, working title, and author list.
- **Step 3: Sequencing Technology:** Indicate Sanger sequencing or Next-Gen platform (Illumina, PacBio, Nanopore).
- **Step 4: Organism & Source Modifiers:** Enter scientific name (e.g., *Escherichia coli*), isolate/strain ID, tissue type, country of isolation.
- **Step 5: Sequence Upload:** Upload FASTA-formatted sequence file. Ensure FASTA header contains matching `[organism=...]` and `[isolate=...]` modifier tags.
- **Step 6: Feature Annotation:** Specify feature intervals (e.g., CDS `54..1420`, gene name, product name). Check for premature stop codons in translation.
- **Step 7: Review & Submit:** Validate warnings and submit to receive a temporary submission tracking number followed by the final Accession Number (e.g., `PQ123456`).

3. Practical Protocol: Protein Data Bank (PDB) Coordinate Parsing

Lab Task 2: Distinguishing & Extracting Coordinates (ATOM vs. HETATM):

Examiners often ask: 'Find the resolution of PDB 1TUP and write down the coordinates of the first bound zinc atom or ligand.'

Key PDB Line Anatomy:

REMARK	2	RESOLUTION.	1.80	ANGSTROMS.						
ATOM	1	N	MET	A	1	27.340	24.430	2.614	1.00	9.67
ATOM	2	CA	MET	A	1	26.266	25.413	2.842	1.00	10.38
HETATM	3294	ZN			401	18.420	15.110	12.330	1.00	15.20
HETATM	3295	O	HOH	A	501	31.220	28.910	4.150	1.00	12.40

- **Resolution Check:** Search for 'REMARK 2' line near the top of the file.
- **Isolating Ligand / Drug / Ion:** Filter records beginning with 'HETATM'. Notice residue name (e.g., 'ZN', 'ATP', 'STI' for Gleevec).
- **Columns Breakdown:** Columns 31–38 = X coordinate; 39–46 = Y coordinate; 47–54 = Z coordinate; 55–60 = Occupancy (\$1.00\$); 61–66 = Temperature / B-factor (\$15.20\$).

4. Practical Protocol: SCOP & CATH Structural Classification Query

How to Look Up a Protein's Structural Domains:

1. Open CATH portal (`cathdb.info`) or SCOPe (`scop.berkeley.edu`).
2. Enter PDB ID (e.g., `1TIM` for Triosephosphate Isomerase).
3. In **CATH**, observe the 4-number classification string (e.g., `3.20.20.10`):
 - **3**: Class = Alpha-Beta.
 - **20**: Architecture = Alpha-Beta Barrel (TIM barrel).
 - **20**: Topology = TIM Barrel.
 - **10**: Homologous Superfamily = Triosephosphate Isomerase-like.
4. In **SCOP**, locate the classification hierarchy: *Class $\rightarrow$ Fold $\rightarrow$ Superfamily $\rightarrow$ Family*.

5. Practical Protocol: Functional Motif Scanning using PROSITE

Lab Task 3: Scanning a Sequence for Motifs via ScanProsite:

1. Navigate to *ScanProsite* (`prosite.expasy.org/scanprosite`).
2. Paste protein sequence or enter UniProt accession (e.g., `P04637`).
3. Choose 'Scan for patterns and profiles'. Click 'Start Scan'.
4. **Interpreting Results:** Distinguish between high-specificity profiles (hit domains like `P53_TAD`, `P53_DNA_BIND`) and short, non-specific patterns (e.g., PKC phosphorylation sites, N-glycosylation sites).

6. Practical Protocol: STRING & BioGRID Network Construction

Lab Task 4: Building a PPI Network & Customizing Confidence Scores:

1. Open STRING (`string-db.org`). Select 'Protein by name'.
2. Enter gene symbol (e.g., `TP53`) and organism (`Homo sapiens`). Click Search.
3. **Interaction Network Display:** Each circle is a protein node; lines (edges) represent predicted/demonstrated interactions.
4. **Setting Confidence Threshold:** Go to 'Settings' $\rightarrow$ Set 'Minimum required interaction score' to **Highest confidence (0.900)** or **Medium (0.400)**.
5. **Turning Evidence Channels On/Off:** Uncheck 'Textmining' to retain only experimentally verified or database-supported interactions.
6. **Functional Enrichment:** Click 'Analysis' tab to view enriched GO terms (Biological Process, Molecular Function) and KEGG pathways with FDR values.

7. Practical Protocol: UCSC Genome Browser & BLAT Search

Lab Task 5: Genomic Localization & BLAT Alignment:

1. Go to `genome.ucsc.edu` $\rightarrow$ Tools $\rightarrow$ **BLAT**.
2. Select Genome: 'Human', Assembly: 'Dec. 2013 (GRCh38/hg38)'.
3. Paste query nucleotide sequence or primer. Click 'Submit'.
4. In the BLAT Results table, select the top hit with 100% identity and click 'browser'.
5. **Navigating the Genome Browser Window:**
 - Chromosome ideogram shows the cytoband (e.g., `17p13.1`).
 - Gene tracks: 'NCBI RefSeq' or 'GENCODE' show exons (solid blocks) and introns (lines with arrows denoting 5' to 3' strand direction).
 - Track configuration: Right click any track to change mode: *Hide, Dense, Squish, Pack, Full*. (Exam tip: 'Pack' or 'Full' shows individual transcript isoforms!).

8. Practical Protocol: Gene Expression Omnibus (GEO) Query

Lab Task 6: Retrieving Microarray Datasets and Exploring Samples:

1. Access NCBI GEO (`ncbi.nlm.nih.gov/geo`).
2. Enter a study accession in the search box: e.g., `GSE19804` (Non-smoking female lung cancer).
3. **Identifying Metadata Components:**
 - **Platforms (GPL):** Identify chip used (e.g., `GPL570 [HG-U133_Plus_2]` Affymetrix Human Genome U133 Plus 2.0 Array).
 - **Samples (GSM):** Inspect sample count (e.g., 120 samples: 60 paired tumor vs normal tissues).
 - **Data Download:** 'Series Matrix File(s)' contains normalized log2 expression values ready for R/Python/Excel analysis; 'RAW' contains `.CEL` files.

TOP 25 VIVA VOCE QUESTIONS & EXAMINER TRAPS

1. What is the difference between an accession number and a version number?

Answer: An Accession Number (e.g., NM_000546) is a permanent, immutable identifier for a record. The Version Number (e.g., NM_000546.6) increments whenever changes or updates are made to the underlying sequence.

2. How does RefSeq differ fundamentally from GenBank?

Answer: GenBank is an archival, redundant database of direct author submissions that can contain duplicates and errors. RefSeq (NCBI) is non-redundant, expertly curated, and provides reference standard sequences (identified by an underscore in accession: NM_, NP_, NC_).

3. What does '/' mean at the end of a GenBank or EMBL flat file?

Answer: It signifies the termination / end-of-record marker for that biological entry.

4. What are the three letters following 'P1;' in a PIR format?

Answer: P1 indicates a protein sequence in PIR/NBRF format. F1 indicates a nucleic acid sequence.

5. Why are somatic mutations cataloged in COSMIC not typically found in dbSNP?

Answer: dbSNP catalogs germline variations and common polymorphisms across populations, whereas COSMIC exclusively catalogs acquired somatic mutations occurring specifically in tumor tissues.

6. What is the significance of the B-factor (temperature factor) in a PDB file?

Answer: B-factor reflects the thermal mobility or static displacement/disorder of that atom in the crystal lattice. Higher B-factors indicate greater flexibility or uncertainty in atomic position.

7. What is the difference between SCOP and CATH in handling multi-domain proteins?

Answer: Both databases split multi-domain proteins into individual structural domain units before classifying each domain independently.

8. In CATH, can two proteins with different topologies share the same Architecture?

Answer: Yes! Architecture only considers the overall 3D shape and orientation of secondary structure elements, regardless of how loops connect them. Topology requires identical connectivity.

9. In a PROSITE pattern, what does '[ST]-x-[RK]' represent?

Answer: Serine or Threonine, followed by any single residue, followed by Arginine or Lysine (a classic substrate recognition motif for certain kinases).

10. What is an EC number for Hexokinase?

Answer: EC 2.7.1.1 (Transferase -> Phosphotransferase -> with an alcohol group as acceptor).

11. What is a Directed Acyclic Graph (DAG) in the context of Gene Ontology?

Answer: A network where nodes (GO terms) are connected by directed edges (e.g., 'is_a', 'part_of') with no closed loops, allowing a specific child term to have multiple parent terms.

12. What are the five main edge colors in STRING networks and their meanings?

Answer: Green: Neighborhood; Red: Fusion; Blue: Co-occurrence; Black: Co-expression; Pink: Experimental data; Light blue: Curated databases; Yellow: Text mining.

13. Why is BioGRID referred to as a curated literature repository?

Answer: Because all recorded physical and genetic interactions in BioGRID are extracted and validated by expert biocurators reading primary peer-reviewed scientific literature.

14. What is the difference between a physical and a genetic interaction in BioGRID?

Answer: A physical interaction means direct binding between proteins (detected via Yeast Two-Hybrid, Co-IP). A genetic interaction means the phenotype of a double mutant deviates from expectation (e.g., synthetic lethality), implying shared pathways without direct contact.

15. What is the primary advantage of BLAT over BLAST in the UCSC Genome Browser?

Answer: BLAT indexes the entire genome into memory using non-overlapping k-mers, making it roughly 500 times faster for finding near-identical matches (>95% similarity) across vertebrate genomes.

16. In UCSC, what is the difference between 'Dense' and 'Pack' track display modes?

Answer: 'Dense' collapses all features into a single solid black horizontal line. 'Pack' displays all transcript isoforms separately with arrows showing strand direction.

17. What is the role of an anchoring enzyme (e.g., NlaIII) in SAGE?

Answer: It cleaves cDNA at a frequent 4-base recognition site (CATG) to ensure all tags are generated from a consistent, defined position near the 3' end of transcripts.

18. In two-color microarrays, what dye is traditionally used for Cy3 and Cy5?

Answer: Cy3 fluoresces Green (~570 nm); Cy5 fluoresces Red (~670 nm).

19. What is a 'probe set' in an Affymetrix GeneChip?

Answer: A group of 11 to 20 probe pairs (25-mers) representing different regions of the same target mRNA transcript.

20. What is a GEO Series Matrix file?

Answer: A pre-formatted, tab-delimited text file containing the normalized expression matrix of all samples (GSMs) in a study, accompanied by sample phenotype descriptions.

21. Can Swiss-Prot accession numbers change?

Answer: Accession numbers in Swiss-Prot are stable and permanent. If entries merge, secondary accession numbers are retained to track provenance.

22. How do you identify the resolution of a crystal structure in a PDB file?

Answer: Look at the `REMARK 2 RESOLUTION. ANGSTROMS.` record near the top of the PDB header.

23. What does 'VUS' mean in ClinVar?

Answer: Variant of Uncertain Significance - a detected variation where current evidence is insufficient to classify it as definitively pathogenic or benign.

24. What is the key functional role of KEGG Orthology (KO)?

Answer: KO links genes across diverse species to an identical functional node in a reference pathway based on sequence homology.

25. Which software is used to build and query Pfam HMM profiles?

Answer: The HMMER software suite (commands: `hmmbuild`, `hmmsearch`, `hmmscan`).

IMPORTANT EXAM TIP: Review the ATOM vs HETATM columns and the CATH vs SCOP hierarchy levels right before stepping into the viva room. These are the two most frequently tested spotting questions!